Pooled Calibration Under Outcome Scarcity Across Intensive Care Hospitals

***Abstract*—Recalibrating a clinical risk model at a hospital with very few deaths is unstable, while pooling calibration data across hospitals may ignore local differences. We study this trade-off for intensive care mortality prediction. A TF-IDF logistic model trained at source hospitals is recalibrated at each target hospital either with a regularized local intercept or with a 12-bin isotonic map fitted to calibration counts, deaths, and score sums summed across hospitals under CKKS homomorphic encryption, emulated on a single host. The prespecified estimand is the hospital-macro difference in held-out Brier score at target hospitals with 0–5 calibration deaths. On eICU (14,110 stays, 45 hospitals), among 11 target hospitals with 0–5 calibration deaths, the local intercept had lower mean Brier score than the pooled map (ΔBrier -12.0×10^{-3}, 95% interval $[-20.0, -4.0]\times10^{-3}$). Against a source-only binned map the interval included zero, and against source Platt scaling with a local intercept the interval included zero. A protocol-fixed simulation running the same code on 160 synthetic 60-hospital cohorts with known risks locates the boundary: with homogeneous prevalence pooling reduced risk error in every replicate, but as heterogeneity grew the share of low-event hospitals harmed rose from 1% to 44%; pooling beat a source-only map only when targets differed systematically from sources. Encrypted and plaintext maps agreed to within 4×10^{-10}.**

***Index Terms*—clinical prediction; calibration; eICU; outcome scarcity; homomorphic encryption; hospital heterogeneity**

I. Introduction

A probability model has two distinct jobs: rank patients by risk and estimate an absolute event probability. Good ranking does not establish that predicted risks agree with observed risks, particularly after movement between hospitals with different case mix, documentation practices, or outcome prevalence [1]. A calibration map fitted on one hospital may be too noisy when few deaths are observed. Combining evidence across hospitals can stabilize a map but can also bias risk estimates at a hospital whose event rate differs from the pooled population.

The eICU Collaborative Research Database includes intensive care records from many United States hospitals [2], [3]. We separate hospitals used to train the predictor from hospitals used to assess its recalibration. We define a 24-hour recorded-entry-time landmark: diagnosis and treatment entries are retained only if recorded from ICU admission through minute 1,440 [4], [5]. This restriction does not establish clinical onset or fully prospective availability. It changes the eligible population: patients discharged from the selected eICU unit before 24 hours are excluded, and each unique patient identifier contributes at most one deterministically selected unit stay. We do not interpret estimates from the earlier, unaudited cohort as evidence for the landmark cohort.

Distributed recalibration itself is established. Huang and colleagues developed privacy-preserving distributed calibration in clinical data, including smooth isotonic regression [6]. FedCal studies federated local and global calibration [7], Cormode and Markov study federated calibration and evaluation of binary classifiers [8], and Maddock and colleagues study private federated post-hoc binning and scaling under data heterogeneity [9]. EquiFL-X uses label-aggregated histogram calibration in federated mammography [10]. The question here is narrower: after strong local and source-only comparators, how does pooled calibration behave at target hospitals with very few local outcome events? We specify a hospital-level probability-accuracy comparison, report calibration diagnostics separately, and report when pooling appears harmful rather than assuming a universal gain.

Homomorphically encrypted post-deployment calibration has also been studied by adjusting a model's last linear layer on image datasets [11]; the present analysis instead fits a post-hoc probability map for clinical text predictors from aggregated hospital statistics.

II. Materials and Methods

A. Study design and data source

This is a retrospective, multi-hospital prediction-model evaluation using eICU Collaborative Research Database version 2.0 [2], [3]. The target is hospital-discharge mortality among patients whose selected eICU unit stay reaches the 24-hour landmark. The frozen loader first requires patient, hospital, unit-visit-number, and unit-discharge-offset fields, then sorts remaining rows by unique patient identifier, unit visit number, and stay identifier and retains one observed unit stay per patient. This pre-selection missing-field exclusion can select a different stay when an earlier ordered row lacks a required field; an aggregate audit that replays the selection rule counts such promoted stays (Table I). Unit visit numbers identify order within a hospital admission, while eICU provides no systematic chronological ordering of a patient's separate hospital stays within the same discharge year [12]. We therefore call this a deterministic selected observed stay, not the first lifetime, hospital, or database ICU stay. Hospital-discharge status is subsequently required, the selected unit-discharge offset must be at least 1,440 minutes, and hospitals must have at least 120 eligible records with more than 20 characters of extracted text. The frozen code flags a nonmissing discharge status containing `expired`, `death`, or `dead` as positive; its agreement with documented status categories requires an aggregate code-frequency audit [12]; an aggregate

diagnostic confirmed that every final-cohort status is `alive` or `expired` and that labels agree with `expired`.

Only diagnosis and treatment entries with offsets from zero through 1,440 minutes are used. These offsets record when an item was entered in eCareManager, so the restriction defines a recorded-entry-time landmark and does not establish clinical onset or fully prospective availability [4], [5]. The admissionDx field is omitted from the primary analysis because the eICU admission-diagnosis documentation permits entry during the first 36 hours and specifies an entered-time offset [13]. A missing diagnosis or treatment offset column raises an error, and entries without a recorded offset are excluded rather than admitted unwindowed. An empty treatment string cannot establish that a treatment was not administered [5]. Text features are built locally; the study does not upload eICU records to a hosted language-model service.

B. Model development and hospital partition

The earlier project notebook specified five hospital-shuffling seeds (7, 11, 17, 23, 31), reused for this corrected analysis, with $\max(2, \lfloor 0.34H \rfloor)$ of H eligible hospitals assigned as source hospitals and the remainder as targets. Within each hospital, about 50% of eligible selected stays are reserved for model development, 25% for calibration, and the remainder for untouched testing, using the seed-specific random partition. Only source-hospital development rows train the base classifier; target-hospital development rows are unused. Calibration rows and their outcome labels from both source and target hospitals contribute to the pooled map. The target hospitals are therefore held out from predictor training, not from calibration. This design evaluates collaborative recalibration after local outcomes have become available; it does not evaluate zero-label transfer to a new hospital. No test rows or test outcomes are used to train the predictor or calibrators. Seed 7 is the designated primary source–target split; the other seeds assess sensitivity and are not independent clinical replications. This analysis was specified before inspection of the corrected run, but was not prospectively registered and followed exploratory work on an earlier cohort.

The model uses a word and bigram TF-IDF vocabulary with sublinear term frequency (at most 30,000 features, minimum document frequency two) fitted on source development text, followed by L2-penalized logistic regression (liblinear, $C = 1$) with balanced class weights. The uncalibrated decision score is converted to a probability by the logistic link. Class weighting can shift probability levels, so improvements relative to raw scores alone are not evidence of an advantage from pooling. No clinical language model is used.

C. Recalibration methods

For source-only temperature scaling, one positive temperature is fitted by bounded minimization of negative log likelihood ($0.05 \leq T \leq 10$) on source-hospital calibration rows; the transformed prediction is sigmoid(z/T) [14], where z is the base-model logit. Pooled temperature scaling sums each hospital's loss at 96 prespecified, equally spaced candidate temperatures from 0.25 to 5.0 and selects the minimizing aggregate loss. The main pooled method divides uncalibrated probabilities into 12 fixed equal-width bins. Each hospital computes, for every bin b, its count n_b, death count d_b, and probability sum s_b. These quantities are summed across hospitals; a weighted isotonic regression then maps the pooled bin means s_b/n_b to pooled event fractions d_b/n_b with weights n_b. This fixed-bin weighted-isotonic calculation is an aggregate-statistic approximation: it need not equal individual-prediction isotonic regression and is not a reproduction of the distributed smooth-isotonic estimator in [6]. Predictions exactly equal to zero belong to the first bin.

The local comparators use the same hospital-specific calibration split. The code also fits per-hospital temperature and unbinned isotonic maps as exploratory comparators; single-class calibration sets may yield degenerate fits. The separately defined local-only fallback requires both classes for temperature fitting and at least three events and nonevents for its isotonic branch. With at least 25 examples, the fallback fits local isotonic regression when both class counts reach three; otherwise it uses local temperature scaling when both classes occur, and leaves the model unchanged for a single-class calibration set. Because that fallback can disadvantage hospitals with zero events, the revised primary comparator instead updates the source-temperature model by a hospital-specific intercept a, an established model-updating form [15], minimizing the hospital's sum of Bernoulli negative log likelihood plus $2a^2$ over -8 <= a <= 8. The fixed ridge coefficient is an implementation choice rather than a universal standard; a sensitivity analysis evaluates coefficients 0.5, 1, 2, 4, and 8 without selecting from target-test outcomes. Label caps of 25, 50, 100, and 200 are exploratory and subsample only the local comparators; the pooled map still uses every hospital's full calibration summary. Use of every available local calibration label is primary.

The prespecified analysis includes a source-only logistic calibration with a positive slope and intercept, followed by a local offset using the same local outcome labels. It also specifies a leave-one-target-hospital-out pooled isotonic map that excludes the evaluated hospital's own calibration summary while retaining contributions from other hospitals. This control is computed on the single host by subtracting the target hospital's plaintext bin summary from the decrypted aggregate; it is an analytic control, not an encrypted protocol, and releasing both inclusive and leave-one-out aggregates in a deployment would expose that hospital's summary by differencing. To separate cross-hospital pooling from map flexibility, it adds source-only binned isotonic and pooled-temperature controls and a fixed local-intercept penalty grid. We report the designated primary contrast even when a control is discordant: a positive primary interval supports only a qualified comparison with the fixed local intercept, while a broader claim across named comparators requires the corresponding controls to support it. The self-including-versus-leave-one-out contrast is a diagnostic because no non-inferiority margin was prespecified.

D. Encryption scope and aggregation audit

The revised notebook encrypts each hospital's aggregate vector with a TenSEAL CKKS public context, homomorphically adds the ciphertexts, and requests decryption of the sum using a distinct private context [16], [17]. The study executes these roles in one process on one authorized data host. This is a cryptographic arithmetic emulation, not a deployment across independent institutions, an implemented trusted third party, or a formal differential privacy guarantee. A proposed deployment would require an independently controlled key custodian and protection against collusion, malicious reporting, and small-group differencing. No encryption claim will be made if the notebook cannot run TenSEAL in the target environment; plaintext fallback is disabled.

The frozen source specifies CKKS polynomial modulus degree 8,192, coefficient moduli of 60/40/40/60 bits, and scale 2^{40}. Per source–target split and backend, it encrypts five short vectors per participating hospital: 96 temperature-loss terms, one calibration count, and three 12-bin vectors for counts, deaths, and predicted-score sums. For H hospitals the aggregation performs $5(H-1)$ ciphertext-vector additions and decrypts five pooled vectors. These are code-level operation counts, not measured runtime, communication cost, or a distributed security evaluation.

The notebook constructs each hospital's count, event, score-sum, and temperature-loss vectors in plaintext inside that same process before encryption; the decryption key is also accessible to the process. Thus the current implementation cannot establish confidentiality against its host or prove a network protocol between hospitals. It tests encrypted summation and subsequent map construction under the CKKS arithmetic assumptions. Decryption reveals pooled bin-level counts, outcomes, and score sums to the key holder. Small pooled bins, a single participating hospital, or differences between overlapping released aggregates may reveal more than intended; restricting released outputs and suppressing small reliability cells are reporting measures, not a formal leakage bound. A deployment claim would additionally need a specified adversary, separated key custody, authenticated input handling, and a release policy for pooled summaries.

CKKS arithmetic is approximate. The frozen source checks that decrypted bin and event counts round within tolerance and rejects impossible counts; rounded zeros define empty bins. Its reported temperature-loss error divides both encrypted and plaintext loss sums by the same decrypted calibration count, so it does not independently check that count. A supplementary seed-7 driver compares the count, normalized loss curve, bin statistics, and held-out map predictions against plaintext on the authorized host. Results are in Table I. Arithmetic fidelity alone does not establish safe aggregate release for every hospital group size.

E. Outcomes and analysis plan

The designated primary estimand is hospital-macro overall probabilistic accuracy, measured by the mean across target hospitals with 0–5 deaths in their full local calibration sets of ΔBrier = Brier(regularized local intercept) - Brier(pooled binned isotonic) on the hospital's untouched test set. Positive values favor the pooled binned-isotonic method against this named comparator. We report the number of contributing target hospitals, both absolute mean Brier scores, the mean and median within-hospital difference, interquartile range, the numbers of hospitals with strictly lower pooled, strictly lower local, and exactly equal Brier scores, and percentile endpoints from 2,000 paired hospital resamples for seed 7. No formal sample-size calculation was performed; the analysis includes all hospitals meeting its fixed eligibility and partition rules and imposes no five-hospital minimum. With one eligible hospital, every resample repeats its score; the endpoints are descriptive, not a between-hospital interval, and we make no superiority claim. With two to four hospitals, percentile intervals are discrete and may have poor coverage; named controls and site-influence checks are essential. No minimum important Brier difference was prespecified; exclusion of zero would establish statistical direction for the named contrast, not clinical materiality. We do not convert the interval into an unadjusted p-value decision. As an exploratory robustness check computed from the frozen hospital pairs, we report how many single-hospital deletions reverse the sign of the mean difference; the leave-one-out means themselves are withheld because, with few hospitals, they reconstruct hospital-level differences. In the primary stratum many test sets may contain few or no deaths; for a test set with no deaths the Brier score equals the mean squared predicted risk, so differences there largely reflect predicted risk levels. Hospital-level scores remain on the authorized host. Repeated hospital appearances across alternate splits are correlated and are not independent observations. A host-side check reconciled each hospital's primary pooled and penalty-2 local scores with the scarcity and penalty tables.

The locked code computes test-set negative log likelihood, 12-bin equal-width expected calibration error (ECE), 12-bin equal-mass adaptive ECE, and area under the receiver operating characteristic curve. A separate pinned-source diagnostic computes descriptive calibration intercepts and slopes, exact-boundary prediction rates, and held-out reliability bins. These diagnostics pool all seed-7 target hospitals rather than the 0–5-death stratum, so they describe the maps overall and cannot by themselves establish calibration in the primary stratum. Brier score combines calibration and discrimination, so a lower Brier score alone does not establish a calibration-only effect [1]. A calibration-specific interpretation requires the diagnostic evidence; ECE is bin dependent and unstable in small event strata. Macro averages treat hospitals equally, while pooled-patient summaries weight large hospitals more and cannot replace the estimand. For the primary stratum itself, an exploratory check reports calibration-in-the-large (intercept with slope fixed at 1) for the pooled, local-intercept, and source-only binned maps. The pooled map is record-weighted, so large calibration hospitals exert more influence even though the primary effect is hospital-macro. The reporting diagnostic withholds counts and event fractions for reliability bins smaller

than 20 stays, adding complementary suppression when only one bin is small; this is no formal privacy guarantee. Primary inference is fixed before the audited test outputs; exploratory diagnostics receive no multiplicity-adjusted significance claim.

The same patients may contribute to different hospital partitions across the five seeds, so split-level estimates are correlated. Site-level confidence intervals summarize between-hospital variation within the designated split; they do not account for the choice of source hospitals or establish external validity at future hospitals. Recalibration changes probability scale, while strictly monotone transformations preserve ranking and isotonic maps may create ties. No clinical risk threshold, treatment recommendation, or safety guarantee is evaluated in the primary analysis.

F. Supplementary simulation with known risks

Because true risks are unobservable in eICU, we fixed a simulation protocol before running it and recorded its checksum. The simulation executes the hash-verified analysis code unmodified, apart from an in-memory return of fitted predictions: the TF-IDF model, 50/25/25 within-hospital split, 0.34 source fraction, split seed 7, 12-bin pooled isotonic map, penalty-2 local intercept, and TenSEAL CKKS aggregation are those of the locked analysis. Each replicate has 60 hospitals with sizes drawn log-normally around 450 (range 130–3,000) and hospital intercepts $a_h = \mathrm{logit}(0.09) + \sigma z_h$, $z_h \sim N(0,1)$. A patient with severity $x \sim N(0,1)$ has true risk $\mathrm{sigmoid}(a_h + 1.5x)$. The text contains synthetic diagnosis tokens whose presence depends on x and on a hospital-specific documentation shift, which creates cross-site drift. We ran 40 replicates at each $\sigma \in \{0, 0.35, 0.7, 1.0\}$. The primary simulation contrast, averaged over target hospitals with 0–5 calibration deaths, is Δrisk = MSE(local intercept, true risk) − MSE(pooled map, true risk) on test rows; positive values favor pooling. We also report the Brier form of the eICU estimand, the leave-one-target-hospital-out and source-only binned-isotonic controls, and the share of low-event hospitals harmed by pooling. Replicate percentiles describe variation across synthetic worlds, not sampling uncertainty for any real hospital. After seeing these results, we added one post hoc scenario, also fixed and checksummed before it ran: with $\sigma = 0.35$, every target hospital's intercept was shifted by $\delta \in \{-0.7, +0.7\}$ relative to source hospitals, with target membership taken in advance from the frozen seed-7 shuffle. It asks whether pooling adds anything beyond a source-only map when targets differ systematically from sources.

III. Results

A. eICU

The audited cohort had 14,110 selected stays from 45 hospitals (12.5% hospital mortality). In seed 7, 11 of 30 target hospitals had 0–5 calibration deaths. In these hospitals the local intercept had lower mean Brier score than the pooled map (ΔBrier -12.0×10^{-3}, 95% interval $[-20.0, -4.0]\times10^{-3}$; Table I). Pooling gave the lower Brier score in 3 hospitals and the local intercept in 8; no single-hospital deletions reversed the sign of the mean. Against the source-only binned map the interval included zero (+2.29 $[-0.03, +5.06]\times10^{-3}$). Of the five prespecified directional controls, 2 favored pooling, 3 were inconclusive, and 0 favored the comparator. A claim beyond the named local-intercept comparison is therefore not made. Excluding each hospital's own summary changed the contrast by -0.13×10^{-3}. Across local-intercept penalties from 0.5 to 8 the contrast ranged from -12.0 to 46.1×10^{-3}. Encrypted and plaintext maps agreed within 1.6×10^{-9} on 2,196 held-out predictions.

B. Simulation with known risks

Table II and Fig. 1 summarize all 160 replicates; three replicates had no hospital in the 0–5-death stratum and contribute no stratum contrast. With homogeneous prevalence ($\sigma = 0$), pooling lowered the stratum-mean risk error relative to the local intercept in every replicate with stratum hospitals (mean 8.7×10^{-3}), and only 1% of low-event hospitals were harmed. At $\sigma = 0.7$ the stratum mean still favored pooling in 97% of replicates, but 22% of low-event hospitals were harmed and the advantage over all target hospitals vanished. At $\sigma = 1.0$ the stratum mean was near zero with a wide replicate range, 44% of low-event hospitals were harmed, and pooling was worse than the local intercept across all target hospitals in every replicate. The Brier contrast pointed the same way but was noisier: at $\sigma = 0$ it favored pooling in 82% of replicates, whereas the risk-error contrast did so in all of them.

Two controls qualify the gain. Excluding the evaluated hospital's own summary changed the scenario-mean stratum risk error by at most 0.3×10^{-3}, so self-inclusion explained little of it. A source-only binned isotonic map came within 0.4–0.9×10^{-3} of the pooled map, so most of the advantage over the local intercept came from the flexible binned map correcting the class-weighted probability scale, not from pooling across hospitals. The post hoc shift scenario separates the two. When target hospitals had lower mortality than sources ($\delta = -0.7$; about ten stratum hospitals per replicate), pooling beat the source-only map by 5.9×10^{-3} [2.5, 10.7] and the local intercept by 14.6×10^{-3} [6.0, 21.6] in every replicate, and 4% of stratum hospitals were harmed. The source-only map was miscalibrated for the shifted targets. In an exploratory paired check within the same 40 replicates, the leave-one-target-hospital-out pooled map also beat the source-only map, by 5.9×10^{-3} [2.5, 10.6] in every replicate, and self-inclusion contributed only 0.04×10^{-3}. The correction therefore came from other target hospitals' summaries, not from each hospital's own labels. When targets had higher mortality ($\delta = +0.7$), 21 of 40 replicates had no stratum hospital. In the 19 eligible replicates only, the pooled advantage over the source-only map was small and inconsistent (1.6×10^{-3}, positive in 11 of 19) and 28% of stratum hospitals were harmed. Across all runs, rounded CKKS counts and deaths matched plaintext exactly, the encrypted and plaintext temperature choices always agreed, the maximum loss-curve error was 1.0×10^{-11}, and the encrypted and plaintext maps

TABLE I

eICU COHORT AND SEED-7 RESULTS. ΔBRIER IS THE NAMED COMPARATOR MINUS THE FIRST-NAMED METHOD, $\times 10^{-3}$, WITH PAIRED HOSPITAL-BOOTSTRAP 95% INTERVAL (2,000 RESAMPLES); POSITIVE FAVORS THE FIRST-NAMED METHOD.

Measure	Result	Detail
Cohort: stays, hospitals, mortality	14,110; 45; 12.5%	15 source, 30 target hospitals (seed 7); 14,110 patients at the 24-h landmark before text and hospital-size filters; 0 stays promoted by missing fields
Primary: pooled vs local intercept, 0–5 deaths	**−12.0 [−20.0, −4.0]**	11 hospitals; mean Brier 0.100 pooled vs 0.118 local; median 11.5, IQR 5.9 to 35.0; favors the comparator
Hospitals pooled lower / local lower / tied	3 / 8 / 0	0 of 11 single-hospital deletions reverse the sign of the mean
Control: pooled vs source-only binned map	+2.29 [−0.03, +5.06]	Tests pooling itself, not map flexibility; inconclusive
Control: pooled vs source Platt + local intercept	−1.29 [−4.80, +2.06]	inconclusive
Control: leave-one-target-out pooled vs local	17.2 [5.4, 28.6]	Excludes the evaluated hospital's own summary; favors pooling
Control: leave-one-target-out pooled vs source Platt + local	−1.43 [−5.07, +2.23]	inconclusive
Control: pooled isotonic vs pooled temperature	90.2 [65.1, 113.1]	favors pooling
Diagnostic: leave-one-out vs self-including pooled	−0.13 [−0.37, +0.05]	Contribution of the hospital's own labels
Local-intercept penalty 0.5, 1, 2, 4, 8	4.8, 9.0, −12.0, 30.2, 46.1	Intervals excluding 0: 4 of 5
Calibration intercept/slope, all target stays	−0.21/0.86	Pooled map; source Platt −0.09/0.96; local intercept −0.78/0.78; ideal 0/1; 30 hospitals
Calibration-in-the-large, 0–5-death stratum	−0.15	Pooled map; local intercept −0.80; source-only binned −0.23; ideal 0; 521 test stays, 60 deaths
CKKS vs plaintext, held-out map predictions	1.6×10^{-9}	Max abs. error over 2,196 predictions; rounded counts and deaths exact; temperature choice matched in all 5 splits

TABLE II

SYNTHETIC SIMULATION, 40 REPLICATES PER σ. CONTRASTS ARE LOCAL MINUS POOLED ($\times 10^{-3}$); POSITIVE FAVORS POOLING.

σ	Hosp.	Δrisk [2.5, 97.5%]	> 0	ΔBrier (> 0)	Harmed	All
0.00	3.3	8.7 [2.7, 17.1]	100%	8.0 (82%)	1%	6.4
0.35	3.6	8.2 [1.4, 15.3]	97%	9.3 (90%)	11%	4.6
0.70	4.7	6.8 [0.5, 15.0]	97%	8.5 (97%)	22%	−0.3
1.00	6.0	0.9 [−16.3, 19.1]	60%	0.7 (55%)	44%	−7.6
$-0.7^{\dagger}$	9.8	14.6 [6.0, 21.6]	100%	14.0 (100%)	4%	9.0
$+0.7^{\dagger}$	0.7	1.9 [−3.7, 7.1]	79%	3.2 (53%)	28%	−0.2

Hosp.: mean number of 0–5-death target hospitals. > 0: share of replicates favoring pooling. Harmed: mean share of those hospitals where pooling increased risk error. All: Δrisk over all target hospitals. †Post hoc: $\sigma = 0.35$ with target-hospital shift δ; for $+0.7$, contrasts use only the 19 of 40 replicates with a stratum hospital.

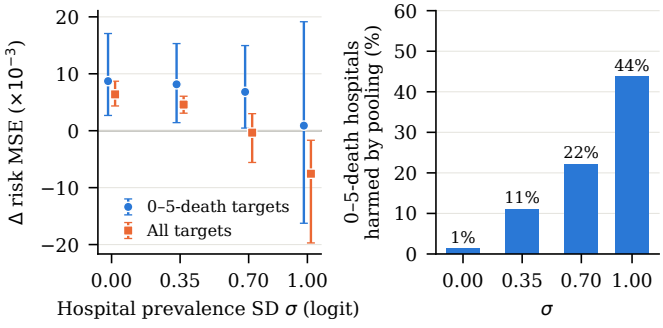


Fig. 1. Simulation boundary. Left: mean Δrisk (local minus pooled) with 2.5–97.5% replicate ranges, for low-event targets and all targets. Right: share of low-event hospitals harmed by pooling.

differed by at most 3.8×10^{-10} over 1,484,326 held-out predictions.

IV. DISCUSSION

This evaluation asks where pooled calibration helps enough to justify applying a shared probability map when outcome labels are scarce. Distributed smooth isotonic regression was already shown by Huang and colleagues [6], and federated approaches to client calibration were developed in FedCal [7] and later work [9]. The contribution is a direct event-stratified test against a local intercept that remains defined even without observed deaths, with controls that separate pooling from map flexibility. In eICU the contrast with the source-only binned map was inconclusive, so these data do not establish that pooling across hospitals adds value beyond a source-only map; the simulation shows that this contrast is small when targets resemble sources. Against source Platt scaling with a local intercept, a stronger local comparator, the interval included zero, so the eICU evidence does not show pooling beating a well-specified local recalibration.

In simulation, pooling helped low-event hospitals when prevalence was similar across hospitals and harmed a growing minority as prevalence diverged; a positive stratum mean coexisted with 22% harmed hospitals at $\sigma = 0.7$. This is the boundary the eICU analysis must locate, and it is why hospital-level harms are reported beside the mean. The simulation also warns that, against this local intercept, a pooled advantage may mostly reflect map flexibility when targets resemble sources. Pooling earned its keep over a source-only map only when targets differed systematically from sources, notably when they had lower mortality, which is the situation that produces low-event hospitals. The source-only binned control is therefore essential in eICU, and its contrast with the pooled map is the direct test of whether cross-hospital pooling adds value. A pooled map absorbs information from target hospitals' calibration rows, but it can also move predictions away from the correct risk at a hospital with a distinctive prevalence.

Thus mean improvements across hospitals cannot establish uniformly better site-specific risk. The apparent gain over uncalibrated predictions may mainly repair the probability shift caused by class weighting; the local-intercept primary comparison and source-only intercept sensitivity are more informative. Event-scarcity strata describe data availability and do not establish why a method wins: sample size, prevalence, score range, and documentation practice may be correlated. The pooled map is patient-weighted; a hospital-equal variant was not evaluated. Proper scoring rules, site-level differences, calibration plots, and a breakdown by event count are needed to interpret the direction and magnitude of any gain.

Several limits remain. The eICU cohort is retrospective and its structured diagnosis and treatment strings are not free-text notes. Excluding early unit discharges changes case mix, and hospital-discharge status is not a time-limited mortality endpoint. The visit-number rule cannot order separate hospital admissions; pre-selection missing-field exclusion can also promote another observed stay and requires sensitivity analysis. Stepdown and other unit labels may affect interpretation. Small hospitals have noisy test estimates, the 120-record threshold is pragmatic, and repeated partitions are not external validation. The primary hospital bootstrap does not refit the pipeline or add within-hospital test-sample uncertainty. Class weighting shifts probability levels, and no unweighted refit isolates its effect. Results are for one designated split. The encryption demonstration does not protect against a malicious keyholder or differencing attacks. The local-intercept penalty is an implementation choice. Reporting is guided by TRIPOD+AI and TRIPOD-Cluster [18], [19]; completed checklists are required before any compliance claim. The study cannot claim improved patient outcomes, clinical utility, differential privacy, or deployed multi-institution security.

Recent eICU work also studies hospital-specific post-hoc adaptation. Jeong and colleagues report a preprint on retrieval adaptation of frozen ICU predictors across hospitals [20]. Their method retrieves outcome information from similar patients; ours examines encrypted pooled calibration summaries and an explicitly low-event local comparator. We therefore do not claim that cross-hospital adaptation itself, or few-label hospital analysis by itself, is new.

V. Conclusions

This source-hospital predictor and separate calibration and test subsets permit evaluation of post-deployment recalibration. On synthetic hospitals with known risks, the unchanged analysis code shows that pooled calibration helps low-event hospitals when prevalence is homogeneous and harms a growing share as prevalence diverges; most of the gain over the local intercept comes from map flexibility unless target hospitals differ systematically from sources, in which case pooling also beats a source-only map; encrypted aggregation matched plaintext to within 4×10^{-10}. On eICU, a regularized local intercept had lower mean Brier score than pooled calibration at low-event hospitals. Whether pooling adds value beyond a source-only binned map was inconclusive.

VI. Data and code availability

eICU Collaborative Research Database version 2.0 is available through PhysioNet to credentialed users under its data-use terms [2]. The companion audited notebook contains the planned analysis but no eICU records; the simulation protocol, harness, and synthetic results contain no patient data; a public code release and any sharing of derived outputs require review of the applicable data-use agreement and author approval.

References

[1] Huang Y, Li W, Macheret F, Gabriel RA, Ohno-Machado L. A tutorial on calibration measurements and calibration models for clinical prediction models. Journal of the American Medical Informatics Association. 2020;27(4):621–633. doi:10.1093/jamia/ocz228.

[2] Pollard T, Johnson A, Raffa J, Celi LA, Badawi O, Mark R. eICU Collaborative Research Database (version 2.0). PhysioNet. 2019. doi:10.13026/C2WM1R. https://physionet.org/content/eicu-crd/2.0/

[3] Pollard TJ, Johnson AEW, Raffa JD, Celi LA, Mark RG, Badawi O. The eICU Collaborative Research Database, a freely available multi-center database for critical care research. Scientific Data. 2018;5:180178. doi:10.1038/sdata.2018.178.

[4] MIT Laboratory for Computational Physiology. eICU Collaborative Research Database: diagnosis table documentation. https://eicu.mit.edu/eicutables/diagnosis/ (accessed 24 September 2026).

[5] MIT Laboratory for Computational Physiology. eICU Collaborative Research Database: treatment table documentation. https://eicu.mit.edu/eicutables/treatment/ (accessed 24 September 2026).

[6] Huang Y, Jiang X, Gabriel RA, Ohno-Machado L. Calibrating predictive model estimates in a distributed network of patient data. Journal of Biomedical Informatics. 2021;117:103758. doi:10.1016/j.jbi.2021.103758.

[7] Peng H, Yu H, Tang X, Li X. FedCal: Achieving Local and Global Calibration in Federated Learning via Aggregated Parameterized Scaler. Proceedings of the 41st International Conference on Machine Learning. PMLR. 2024;235:40331–40346. https://proceedings.mlr.press/v235/peng24g.html

[8] Cormode G, Markov IL. Federated Calibration and Evaluation of Binary Classifiers. Proceedings of the VLDB Endowment. 2023;16(11):3253-3265. doi:10.14778/3611479.3611523.

[9] Maddock S, Cormode G, Maple C. Private Federated Multiclass Post-hoc Calibration. arXiv preprint arXiv:2510.01987. 2025. https://arxiv.org/abs/2510.01987

[10] Öney B. EquiFL-X: Communication-constrained federated calibration via label-aggregated histograms for multi-view screening mammography under acquisition shift. Comput Methods Programs Biomed. 2026;286:109596. doi:10.1016/j.cmpb.2026.109596.

[11] Pathan SMK, Wang Q, Takeshita J, Imran SB, Shetty S. Privacy-Preserving Post Deployment Model Calibration with Fully Homomorphic Encryption. 2026 International Conference on Computing, Networking and Communications (ICNC). IEEE; 2026:1–6. doi:10.1109/ICNC68183.2026.11416991.

[12] MIT Laboratory for Computational Physiology. eICU Collaborative Research Database: patient table documentation. https://eicu.mit.edu/eicutables/patient/ (accessed 24 September 2026).

[13] MIT Laboratory for Computational Physiology. eICU Collaborative Research Database: admissionDx table documentation. https://eicu.mit.edu/eicutables/admissiondx/ (accessed 24 September 2026).

[14] Guo C, Pleiss G, Sun Y, Weinberger KQ. On Calibration of Modern Neural Networks. Proceedings of the 34th International Conference on Machine Learning. PMLR. 2017;70:1321-1330. https://proceedings.mlr.press/v70/guo17a.html

[15] Steyerberg EW, Borsboom GJJM, van Houwelingen HC, Eijkemans MJC, Habbema JDF. Validation and updating of predictive logistic regression models: a study on sample size and shrinkage. Statistics in Medicine. 2004;23(16):2567-2586. doi:10.1002/sim.1844.

[16] Benaissa A, Retiat B, Cebere B, Belfedhal AE. TenSEAL: A Library for Encrypted Tensor Operations Using Homomorphic Encryption. arXiv preprint arXiv:2104.03152. 2021. https://arxiv.org/abs/2104.03152

[17] Cheon JH, Kim A, Kim M, Song Y. Homomorphic Encryption for Arithmetic of Approximate Numbers. Advances in Cryptology - ASIACRYPT 2017. Lecture Notes in Computer Science 10624. Springer; 2017:409-437. doi:10.1007/978-3-319-70694-8_15.

[18] Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Van Calster B, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ. 2024;385:e078378. doi:10.1136/bmj-2023-078378.

[19] Debray TPA, Collins GS, Riley RD, Snell KIE, Van Calster B, Reitsma JB, Moons KGM. Transparent reporting of multivariable prediction models developed or validated using clustered data: TRIPOD-Cluster checklist. BMJ. 2023;380:e071018. doi:10.1136/bmj-2022-071018.

[20] Jeong I, Lee T, Kim B, Park J-H, Kim Y, Lee H. PRAM: post-hoc retrieval augmentation adapts intensive care prediction models without retraining and can be shared across hospitals. medRxiv preprint. 2026. doi:10.64898/2026.04.03.26350132. https://www.medrxiv.org/content/10.64898/2026.04.03.26350132v2